

# Spatial Autocorrelation and Regression Analysis of Global Terrorism Incidents

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**Abstract**—This study investigates the spatial structure of global terrorism incidents using country-level aggregation of the Global Terrorism Database (GTD). The objective is to examine whether terrorist activity exhibits statistically significant spatial clustering and to assess the presence of spatial spillover effects. Country-level attack counts and fatalities were aggregated and merged with administrative boundary shapefiles to construct a spatial weights matrix based on queen contiguity. Global Moran's I was computed to test overall spatial autocorrelation, followed by Local Indicators of Spatial Association (LISA) to identify hotspot and coldspot regions. Results indicate significant positive spatial autocorrelation (Moran's  $I = 0.133$ ,  $p < 0.05$ ), suggesting that countries with high levels of terrorist activity tend to cluster geographically. LISA analysis reveals concentrated high-high clusters in specific regions, alongside large low-low clusters. An ordinary least squares (OLS) regression model was estimated using fatalities as a predictor of attack counts, and residual spatial autocorrelation was detected. A spatial lag model was subsequently fitted, revealing modest and marginally significant spatial spillover effects. Overall, the findings confirm that terrorism exhibits regional clustering, while spatial diffusion effects appear limited after accounting for internal country-level severity measures. The code is available at [1].

## I. INTRODUCTION

Terrorism is not uniformly distributed across the globe; instead, it tends to exhibit strong regional concentrations influenced by political instability, conflict spillovers, and geographic proximity. Understanding whether terrorist activity displays systematic spatial patterns is essential for identifying regional clusters, assessing potential spillover effects, and informing policy interventions. Traditional descriptive statistics can reveal frequency patterns, but they do not explicitly account for spatial dependence between neighboring countries.

Spatial statistical methods provide a formal framework to test whether observed geographic clustering is statistically significant rather than random. Measures such as Moran's I allow for the detection of global spatial autocorrelation, while Local Indicators of Spatial Association (LISA) enable the identification of localized hotspots and coldspots. Furthermore, spatial regression models extend classical regression analysis by incorporating spatial dependence structures, allowing researchers to examine whether outcomes in one region are influenced by neighboring regions.

This study conducts a country-level spatial analysis of terrorism incidents using the Global Terrorism Database (GTD). The objectives are threefold: (1) to evaluate whether terrorist

attack counts exhibit significant global spatial autocorrelation, (2) to identify localized clusters and spatial outliers, and (3) to assess the presence of spatial spillover effects through spatial regression modeling. By combining spatial autocorrelation measures with regression-based analysis, this work provides a systematic examination of the spatial structure underlying global terrorism patterns.

## II. DATASET DESCRIPTION AND PREPROCESSING

The analysis is based on the Global Terrorism Database (GTD) [2], an open-source dataset that documents terrorist incidents worldwide. The dataset contains detailed event-level information, including date, location, attack type, perpetrator group, casualties, and weapon characteristics. For the purposes of spatial analysis, the data were aggregated at the country level to construct areal units suitable for spatial statistical modeling.

From the event-level records, country-wise summaries were computed to obtain the total number of terrorist attacks and total casualties. The primary dependent variable used in subsequent modeling is the total number of attacks per country. In addition, a severity measure was constructed by summing the number of killed and wounded individuals to compute total fatalities. Countries with no recorded terrorist incidents were retained in the dataset to preserve spatial structure, and missing values arising from aggregation were explicitly treated as zeros.

To enable spatial analysis, country-level aggregates were merged with administrative boundary shapefiles obtained from Natural Earth. These polygon boundaries define the spatial units required for constructing neighborhood relationships. A queen contiguity-based spatial weights matrix was generated to represent adjacency between countries sharing common borders or vertices. The weights matrix was row-standardized to ensure comparability across units with varying numbers of neighbors.

The final dataset consists of 242 countries with corresponding attack counts, fatalities, and associated spatial geometry. This preprocessing pipeline ensures that the data are suitable for spatial autocorrelation analysis and spatial regression modeling. Figure 1 shows the spatial distribution of fatalities, which exhibits patterns broadly consistent with attack counts and motivates its inclusion as an explanatory

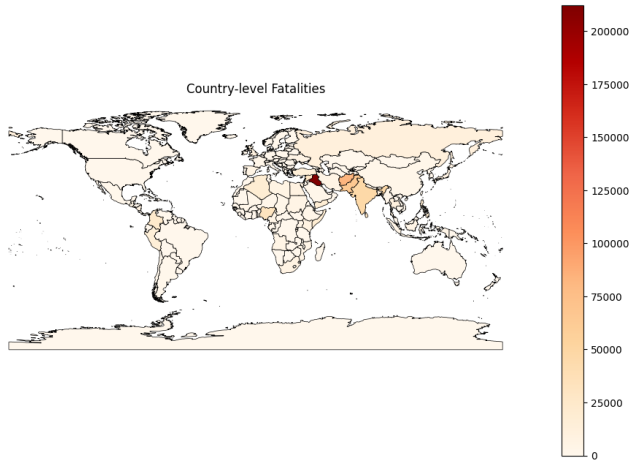


Fig. 1. Country-level distribution of total fatalities resulting from terrorist incidents.

variable in regression analysis. The GTD dataset is particularly suitable for spatial statistical analysis because it provides geocoded event-level information that can be aggregated into well-defined areal units. The presence of country identifiers enables consistent spatial aggregation, while the availability of quantitative measures such as attack counts and fatalities allows application of spatial autocorrelation statistics and regression modeling. Furthermore, the large number of spatial units (242 countries) ensures sufficient variability for reliable estimation of global and local spatial statistics.

### III. SPATIAL METHODOLOGY

#### A. Spatial Weights Construction

Spatial analysis requires an explicit representation of neighborhood relationships among spatial units. In this study, a queen contiguity criterion was used to define spatial adjacency, whereby two countries are considered neighbors if they share a common boundary or vertex. This approach captures direct geographic proximity and is commonly employed in areal spatial analysis. The resulting spatial weights matrix was row-standardized so that the weights associated with each country sum to one. Countries without contiguous land borders (islands) were retained as spatial isolates and treated accordingly in subsequent analysis.

#### B. Global Spatial Autocorrelation

To evaluate whether terrorist attack counts exhibit systematic spatial clustering, Global Moran's I [3] was computed. Moran's I measures the degree to which similar values cluster in space by comparing deviations from the mean across neighboring units. Formally, it assesses whether high (or low) values of a variable tend to occur near other high (or low) values more frequently than would be expected under spatial randomness. Statistical significance was assessed using permutation-based inference, enabling robust evaluation of whether observed clustering deviates from a random spatial distribution.

#### C. Local Spatial Autocorrelation (LISA)

While Global Moran's I provides an overall measure of spatial dependence, it does not identify the specific locations contributing to clustering. Therefore, Local Indicators of Spatial Association (LISA) [4] were computed to detect localized spatial patterns. LISA decomposes global spatial autocorrelation into country-level statistics and classifies each unit into one of four categories: high-high (hotspot), low-low (coldspot), high-low (spatial outlier), and low-high (spatial outlier). Statistical significance was determined using permutation tests at the 5% level. This analysis enables identification of regional clusters and spatial anomalies in terrorist activity.

#### D. Spatial Regression Modeling

To examine whether spatial dependence persists after accounting for explanatory variables, regression analysis was conducted. An ordinary least squares (OLS) model was first estimated with total attack counts as the dependent variable and total fatalities as the explanatory variable. Residual spatial autocorrelation was then tested using Moran's I to determine whether spatial dependence remained unaccounted for.

Given the presence of significant residual spatial autocorrelation, a spatial lag model was estimated. The spatial lag specification incorporates a spatially lagged dependent variable, allowing the outcome in one country to depend partially on outcomes in neighboring countries. The model takes the form:

$$Y = \rho WY + X\beta + \epsilon, \quad (1)$$

where  $Y$  represents attack counts,  $W$  is the spatial weights matrix,  $\rho$  is the spatial autoregressive coefficient capturing spillover effects,  $X$  denotes explanatory variables, and  $\epsilon$  is the error term. The spatial lag model enables assessment of both direct effects and indirect (spillover) effects arising from geographic proximity.

### IV. EXPERIMENTAL RESULTS

#### A. Global Spatial Autocorrelation

Figure 2 illustrates the geographic distribution of terrorist attack counts, revealing visible regional concentrations consistent with the significant Global Moran's I result. Global Moran's I was computed to evaluate overall spatial clustering in country-level terrorist attack counts. The analysis yielded a Moran's I value of 0.133 with a permutation-based p-value of 0.027. The positive and statistically significant result indicates the presence of spatial clustering, suggesting that countries with high levels of terrorist activity tend to be geographically proximate to other high-activity countries. This finding rejects the null hypothesis of spatial randomness and confirms that terrorism exhibits a non-random spatial structure at the country level.

The Moran scatterplot shown in Figure 3 further illustrates the positive relationship between standardized attack counts and their spatial lag, reinforcing the evidence of global spatial dependence.

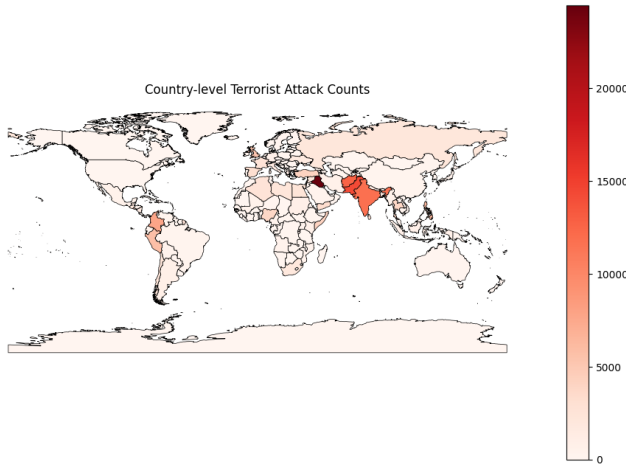


Fig. 2. Country-level distribution of terrorist attack counts. Darker shades indicate higher attack frequencies.

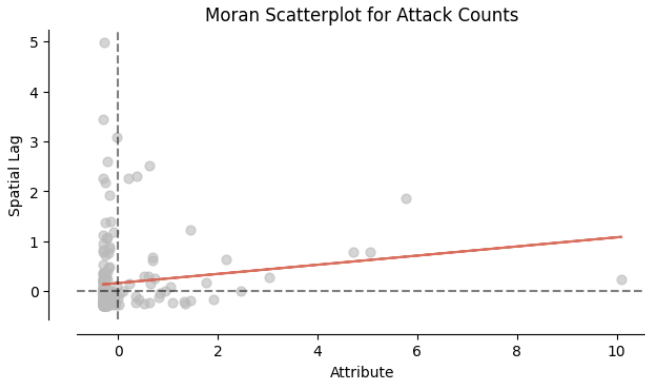


Fig. 3. Moran scatterplot for country-level attack counts. The positive slope indicates positive spatial autocorrelation.

### B. Local Spatial Autocorrelation (LISA)

To identify localized spatial patterns, LISA statistics were computed using a 5% significance threshold. The results indicate the presence of 7 high-high clusters (hotspots), 74 low-low clusters (coldspots), 2 high-low spatial outliers, and 13 low-high spatial outliers, with the remaining countries classified as not statistically significant.

The high-high clusters correspond to geographically concentrated regions of elevated terrorist activity, while low-low clusters represent relatively stable regions surrounded by similarly low-activity neighbors. The presence of high-low and low-high outliers highlights localized anomalies where a country's activity level differs substantially from that of its neighbors. These findings reinforce the conclusion that terrorism is regionally concentrated rather than uniformly distributed. Figure 4 presents the LISA cluster map, highlighting statistically significant high-high (hotspot) and low-low (coldspot) regions, along with spatial outliers.

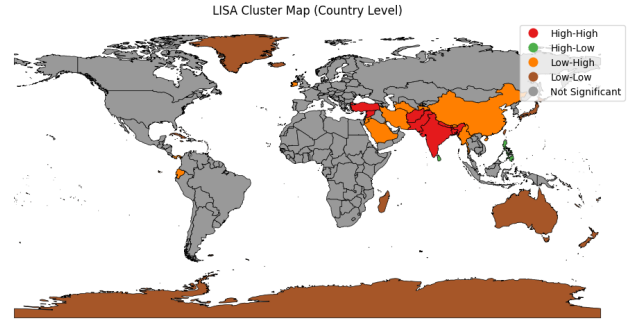


Fig. 4. LISA cluster map identifying statistically significant spatial clusters at the country level.

### C. Ordinary Least Squares (OLS) Regression

An OLS regression model was estimated with total attack counts as the dependent variable and total fatalities as the explanatory variable. The model achieved an  $R^2$  of 0.86, indicating that fatalities strongly explain variation in attack counts across countries. The fatalities coefficient was positive and statistically significant ( $p < 0.001$ ), suggesting that countries experiencing higher casualty counts also tend to record higher numbers of attacks.

However, Moran's I computed on the OLS residuals yielded a value of 0.188 with a p-value of 0.009, indicating significant residual spatial autocorrelation. This result suggests that spatial dependence remains even after controlling for fatalities, implying that OLS may be misspecified in the presence of spatial structure.

To evaluate predictive performance, an 80–20 holdout validation strategy was employed. The dataset was randomly split into training and testing subsets, with the OLS model estimated on the training data and predictions generated for the unseen test data. The resulting test root mean squared error (RMSE) was 745.04. This indicates that while fatalities explain a substantial portion of variation in attack counts, prediction error remains non-trivial, reflecting the high variance and skewness inherent in terrorism incident data.

The spatial distribution of OLS residuals, shown in Figure 5, suggests remaining spatial structure, which is confirmed by the significant residual Moran's I statistic.

### D. Spatial Lag Model

To account for spatial dependence, a spatial lag model was estimated. The spatial autoregressive coefficient ( $\rho$ ) was estimated at 0.0598 with a p-value of 0.096. While positive, the coefficient is only marginally significant at the 10% level and not significant at the 5% level. The spatial lag model achieved a pseudo- $R^2$  of 0.8619, representing only a slight improvement over the OLS model.

Impact decomposition indicates that the direct effect of fatalities remains dominant, while indirect (spillover) effects are comparatively small. These results suggest that although global clustering exists, cross-border spillover effects are mod-

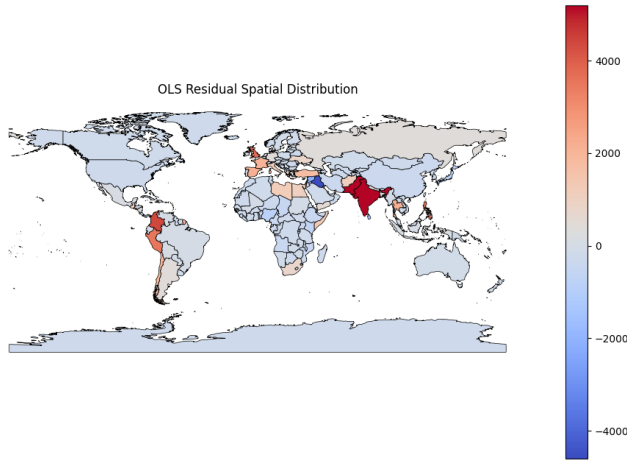


Fig. 5. Spatial distribution of OLS residuals. Residual clustering indicates remaining spatial dependence.

est once internal country-level severity measures are accounted for.

## V. DISCUSSION

The results demonstrate that terrorist activity exhibits statistically significant spatial clustering at the country level. The positive Global Moran's  $I$  confirms that countries with high attack counts tend to be geographically proximate to other high-activity countries. This aligns with the broader understanding that terrorism is often concentrated in conflict-prone regions characterized by political instability, cross-border insurgencies, and shared geopolitical dynamics.

The LISA results further refine this insight by revealing localized high-high and low-low clusters. High-high clusters indicate regional concentrations of elevated activity, suggesting that instability is not isolated but regionally embedded. Conversely, large low-low clusters highlight extensive regions of relative stability. The presence of a small number of spatial outliers (high-low and low-high) suggests that while regional patterns dominate, localized anomalies exist where national conditions diverge from neighboring trends.

The regression analysis provides additional nuance. Although fatalities strongly explain variation in attack counts, residual spatial autocorrelation in the OLS model indicates that purely non-spatial regression is insufficient. However, the spatial lag model reveals only modest and marginally significant spillover effects. This suggests that while terrorism clusters geographically, the diffusion of attack intensity across borders may be limited once internal severity measures are accounted for. In other words, regional clustering appears to be driven more by shared structural conditions than by direct spatial contagion effects.

Several limitations should be acknowledged. First, the analysis is conducted at the country level, which may obscure substantial subnational variation. Second, the regression model includes only a single explanatory variable, limiting the ability to isolate structural determinants of terrorism. Third, the study

does not incorporate temporal dynamics, which may influence patterns of diffusion over time. Future research could extend this framework by incorporating socio-economic covariates, multi-scale spatial analysis, or spatio-temporal modeling approaches.

Overall, the findings highlight the importance of incorporating spatial structure into terrorism research, while also cautioning against overinterpreting spatial dependence as direct cross-border diffusion.

## VI. CONCLUSION

This study examined the spatial structure of global terrorism incidents using country-level aggregation of the Global Terrorism Database. Global Moran's  $I$  revealed statistically significant positive spatial autocorrelation, confirming that terrorist activity is not randomly distributed across countries but instead exhibits regional clustering. Local spatial analysis identified distinct hotspots and coldspots, further demonstrating that terrorism is geographically concentrated in specific regions.

Regression analysis showed that fatalities strongly explain variation in attack counts; however, residual spatial autocorrelation indicated that non-spatial models fail to fully capture geographic dependence. The spatial lag model accounted for spatial structure but revealed only modest and marginally significant spillover effects. These findings suggest that while terrorism clusters regionally, cross-border diffusion effects are limited once internal severity measures are considered.

Overall, the results highlight the value of spatial statistical methods in uncovering geographic structure in terrorism data. Incorporating spatial autocorrelation and spatial regression techniques provides a more rigorous understanding of regional concentration patterns. Future work could extend this analysis by integrating additional socio-economic predictors, exploring subnational spatial scales, or incorporating temporal dynamics to better capture evolving patterns of terrorist activity.

## DECLARATION OF GENERATIVE AI TOOLS

Generative AI tools were used during the preparation of this assignment to assist with code debugging, structural organization of the report, and refinement of written explanations. All spatial modeling procedures, implementation decisions, and interpretations of results were independently executed and verified by the authors. The final analysis, figures, and conclusions reflect the authors' understanding of spatial statistical methods and their application to the Global Terrorism Database.

## TASK OWNERSHIP DECLARATION

This assignment was completed collaboratively by the three authors. The contributions are distributed as follows:

**Keshav:** Led the data preprocessing and spatial analysis pipeline, including aggregation of the Global Terrorism Database at the country level, integration of administrative boundary shapefiles [5], construction of the spatial weights matrix, and implementation of Global Moran's  $I$  and LISA

analysis. Keshav also coordinated the overall analytical workflow and figure generation.

**Rishit:** Implemented and evaluated the regression models, including the OLS baseline model, residual spatial autocorrelation testing, and estimation of the spatial lag model. Rishit contributed to validation of model outputs and interpretation of regression results.

**Himanshu:** Contributed to methodological documentation, interpretation of spatial findings, and refinement of analytical reasoning. Himanshu also assisted in validating results and ensuring consistency between modeling outputs and written explanations.

All authors collaboratively contributed to drafting, editing, and finalizing the report.

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